



MIT SLOAN SCHOOL OF MANAGEMENT  
MIT SCHWARZMAN COLLEGE OF COMPUTING



**MODULE 2 UNIT 1**  
**Lesson Video 1 Transcript**

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### Agent architecture

JACOB ANDREAS: So, our agent has had to work pretty hard to get the user to share all of the relevant information. And then it had to work pretty hard to figure out what the right way to resolve this particular mistaken product situation was, given everything that happened in this user's exceptional case. Now, unless we're careful about how we design this agent, the next time somebody else writes in with the same issue, or maybe even the next time this user comes back asking about the status of the order, the agent is going to have to do all of this work again to figure out how to resolve this user's particular case.

So, one of the most important ways in which we can improve upon this agent architecture that we've laid out so far is giving our agents the ability to learn over time. One straightforward option here would be to keep all of the chats that the agent has with all of the users it ever talks to as part of its input or part of its prompt, so that it can look back at what it did before and refer to these previous conversations.

Now this is probably fine if we know our agent is only ever going to interact with a single user. But if we want to aggregate together information from multiple users, we're going to wind up right back where we started with the sort of big mega prompt that contained all of the information about our entire customer knowledge base. There's simply too much context and too great a possibility that we're going to leak information between users that we don't want this agent to share.

### Memory architecture

ANDREAS: The active area of research right now is aimed at coming up with various kinds of memory architectures for language model agents. We could teach a whole semester long course on this, but the basic idea is to design some way of extracting the useful pieces of information from some single user interaction, storing those useful pieces of information in some sort of compressed form, and maybe anonymous form so that they're easy to access later, and can be shared across conversations.

One thing that we can do is use memory or learning of this kind to build models of users. For example, by building up aggregated profiles of everything that we know about any individual user that we can use the next time we talk to them. For example, remembering things about their preferred style of conversation and the answers to questions that they're likely to ask about their particular order.

We can also use this not to aggregate information about users, but instead about procedures. So, if we think about the example of our agent learning the sort of right policy for responding to this gift return outside of 30 days, we can imagine taking that whole sequence of search results that we ran, storing them in some sort of external knowledge base, so that the next time a user asks a language model agent the same kind of question, rather than having to run this long sequence of searches again, it can simply refer to this answer that it cached that tells you directly how you're supposed to respond to gift returns outside the normal window.

### Learning via updates

ANDREAS: The most powerful thing that you can do is actually take these experiences and use them to update the weights of the language model itself by increasing the probability that it will reproduce successful interactions, and decreasing the probability that it will reproduce unsuccessful interactions. For example, using reinforcement learning algorithms. These kinds of procedures require a lot of data and are computationally very intensive.

Most end users are not going to run reinforcement learning on their own, but providers of large language models certainly do this themselves and sometimes make available services for fine-tuning other models to do so. And so, what we've done so far now is we've taken the agent architecture that we've been building and extended these agents themselves with the ability to learn on the fly, to remember things that they learned about individual users, or about the results of successful interactions with users and the procedures that they use to obtain those results so that they can obtain the same kinds of results or have the same kinds of conversations without needing to do quite so much learning or reasoning in the future.

SPEAKER: Did you understand all the concepts covered in this video? If you'd like to review any of the content, please click on the chapter menu.